

Customer Segmentation Using RFM Analysis and K-Means Clustering: Evidence from an Online Retail Dataset

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Abstract

This study examined whether purchase frequency and monetary value represent a single underlying dimension of customer purchasing behavior or two distinct dimensions that, together, reveal meaningful customer segments. Using transaction data from a UK-based online retailer comprising 4,338 customers, Recency, Frequency, and Monetary (RFM) variables were constructed for each customer. Spearman correlation analysis revealed a strong positive relationship between Frequency and Monetary value ($\rho = 0.807$, $p < .001$), supporting the view that these variables are closely related overall. However, k-means clustering ($k = 4$, validated via the elbow method and silhouette analysis) identified four behaviorally distinct segments — rare buyers, regular repeat customers, loyal high-value customers, and extreme outliers — revealing differences in customer value that a single-dimension view would obscure. Principal component analysis further showed that these segments exist along a continuous behavioral spectrum rather than as sharply bounded groups. Together, these findings suggest that while Frequency and Monetary value are strongly correlated in the aggregate, analyzing them jointly reveals meaningful behavioral heterogeneity that would be overlooked by relying on either measure in isolation, with direct implications for customer segmentation and targeted marketing strategies.

1. Introduction

Customer segmentation is a foundational task in marketing analytics because it enables organizations to better understand customer behavior, allocate marketing resources efficiently, and design targeted retention and acquisition strategies. As businesses continue to generate large volumes of transactional data, converting that data into actionable consumer insight has become an increasingly important analytical challenge. Firms that treat all customers identically risk misallocating resources: over-investing in low-value, one-time buyers while under-serving the smaller group of high-value repeat customers who disproportionately drive revenue. This project addresses that challenge directly by applying Recency-Frequency-Monetary (RFM) analysis in combination with k-means clustering to a publicly available, real-world online retail transaction dataset, with the goal of producing customer segments that are both statistically defensible and practically interpretable.

The central research question guiding this work is: can transaction-level purchase data be used to identify distinct, meaningful customer segments that reflect differences in purchasing behavior? A closely related question follows from it: does the empirical relationship between purchase frequency and monetary value — often assumed within RFM-based segmentation frameworks but rarely tested as a research question in its own right — hold up when examined directly?

Answering these questions matters for two reasons:

- Managerially, well-constructed segments allow firms to tailor retention strategies, promotional targeting, and customer lifetime value estimates to the customers who matter most.
- Methodologically, this project offers an opportunity to examine an assumption commonly built into RFM segmentation — that frequency and monetary value move together — through formal statistical testing rather than treating it as given.

This report documents the full analytical pipeline: from raw transaction data through cleaning, RFM construction, correlation analysis, and k-means clustering, to a final set of four interpretable customer segments. Along the way, it evaluates the appropriateness of the clustering solution using silhouette analysis and visualizes the segment structure using principal component analysis, situating the findings within the broader academic literature on market segmentation and RFM methodology. Beyond producing customer segments, this study emphasizes methodological transparency by documenting each stage of the analytical workflow — from data cleaning through final segmentation — highlighting reproducibility as one of the project's core strengths.

2. Literature Review

The practice of dividing a customer base into meaningful, differentiated groups — rather than treating it as homogeneous — traces back to Smith (1956), whose introduction of market segmentation as a deliberate marketing

strategy, distinct from simple product differentiation, remains foundational to the logic underlying this study's approach. Smith's central insight, that heterogeneous demand within a market can be more effectively served by identifying and targeting distinct subgroups than by a single undifferentiated offering, continues to underpin nearly all subsequent segmentation research. Since Smith's original conceptualization, however, advances in information systems and transactional databases have shifted segmentation practice away from demographic and psychographic approaches and toward behavioral methods that rely directly on observed purchasing activity — a shift that leads naturally into the RFM framework used in this study.

The RFM framework became widely established through research on direct marketing, where Bult and Wansbeek (1995) demonstrated how transactional purchasing histories could be used to identify profitable customers more effectively than undifferentiated marketing approaches. RFM has since become a standard framework in customer relationship management (CRM) for identifying and prioritizing valuable customers using behavioral rather than demographic data.

This study's underlying motivation — understanding customer value in order to inform differentiated marketing and retention strategy — is further grounded in customer lifetime value (CLV) theory, notably Reinartz and Kumar (2000), who demonstrated empirically that customer profitability is not uniform across a customer base, and that long-standing customers are not automatically the most profitable. This finding highlights the importance of empirically driven segmentation rather than assuming that loyalty and customer value always coincide.

Turning to method, the clustering technique applied here, k-means, was originally introduced by MacQueen (1967) as a method for partitioning multivariate observations into groups that minimize within-group variance. K-means has since been widely applied specifically to RFM-based customer segmentation in retail and e-commerce contexts. Christy, Umamakeswari, Priyatharsini, and Neyaa (2021) combined RFM ranking with k-means and fuzzy c-means clustering to identify key customer segments, demonstrating the framework's flexibility across clustering algorithms. More directly relevant to the present study, John, Shobayo, and Ogunleye (2023) applied and compared several clustering algorithms, including k-means, to the same UK online retail dataset used in this project, providing a useful point of methodological comparison for the results reported here.

Taken together, this literature establishes both the theoretical rationale for segmentation (Smith, 1956), the specific behavioral framework used to operationalize it (Bult & Wansbeek, 1995; Reinartz & Kumar, 2000), and the clustering methodology used to derive segments empirically (MacQueen, 1967; Christy et al., 2021; John et al., 2023). Unlike many applied RFM studies, which focus primarily on producing customer segments, the present study integrates formal hypothesis testing with methodological validation: this approach not only identifies interpretable customer segments, but also evaluates the statistical relationship among RFM dimensions and the robustness of the resulting segmentation, while incorporating multiple validation and visualization techniques — including silhouette analysis and principal component analysis — to strengthen the interpretability and reproducibility of the results.

3. Methodology

3.1 Dataset

This study uses the "Online Retail II" dataset, a publicly available transactional dataset originally made available through the UCI Machine Learning Repository and commonly accessed today through data science platforms such as Kaggle. The dataset contains transaction-level records from a UK-based online retailer specializing in unique all-occasion giftware, covering the period from December 2009 to December 2011. Each row represents a single line item within an invoice and includes an invoice number, product code, product description, quantity purchased, invoice date and time, unit price, customer ID, and customer country.

The raw dataset contains 541,910 transaction records. This dataset is well suited to the present study because it records individual purchasing activity at the transaction level, with a unique customer identifier attached to each record — making it possible to reconstruct each customer's recency, frequency, and monetary value directly from observed behavior, rather than from self-reported or demographic proxies. This granularity is precisely what the research question requires: a behavioral basis for identifying meaningful customer segments. Because the analysis relies on constructing these customer-level measures, each transaction record must be reliably attributable to an individual customer, and each recorded purchase must represent a genuine, completed transaction. The data preparation steps required to meet these conditions are described in the following section.

3.2 Data Preparation

Initial inspection of the raw dataset — using summary statistics and structural checks — revealed three primary data quality issues requiring resolution before the analysis could proceed: negative values in the Quantity field, negative and zero values in the Price field, and a substantial number of missing Customer ID values (135,080 rows, roughly 25% of the dataset). Each issue was addressed through a series of data preparation procedures, guided by the principle that only records reflecting genuine, completed, individually attributable purchases should inform the construction of customer-level behavioral measures.

First, rows whose invoice number began with the letter "C" — the dataset's coding convention for cancelled transactions, corresponding to the negative Quantity values observed during inspection — were removed. A cancelled order does not reflect a completed purchase decision and cannot meaningfully inform an analysis of purchasing behavior. Second, a small number of rows contained zero or negative prices; inspection indicated these were not genuine consumer transactions but bookkeeping artifacts, including manual "bad debt" adjustment entries recorded internally by the company. These rows were removed, as they represent internal accounting adjustments rather than consumer purchase decisions, and their inclusion would have misrepresented the construct under study.

Third, approximately 25% of rows had no associated Customer ID. These records likely correspond to guest checkouts, in which a purchase was completed without an account being created. Because customer-level segmentation requires

transactions to be linked to individual customers, these records were excluded from the analysis; the Customer ID column itself was retained, with only the incomplete rows removed.

Finally, Price contained an extreme high value (maximum = 38,970) relative to the bulk of the data. Rather than being removed, this observation was retained because no evidence suggested it represented a recurring error; its potential influence was addressed analytically through the use of rank-based correlation and variable scaling prior to clustering, rather than through arbitrary deletion. The InvoiceDate field, originally read in as a character/text field, was also converted to a proper date-time format, a necessary step for computing recency and for correctly ordering transactions over time.

Taken together, these data preparation procedures reduced the dataset from 541,910 rows to 397,885 rows, a removal of 144,025 rows (26.6%) across all filtering steps combined. Table 1 summarizes this reduction; per-step row counts were not individually recorded during the original filtering process, and are therefore not reported here rather than reconstructed retroactively. The full cleaning code, provided in Appendix B, allows this process to be reproduced exactly.

Table 1. Data Cleaning Summary

Stage	Number of Records
Raw dataset	541,910
Cleaned dataset (after all filtering steps)	397,885
Total records removed	144,025 (26.6%)

3.3 Construction of the RFM Variables

Following the Recency-Frequency-Monetary (RFM) framework described in the Literature Review, each customer's total spend on a given transaction line was first computed as the product of Quantity and Price. These line-level totals were then aggregated by Customer ID to produce a single row per customer, summarizing three behavioral measures. Recency was calculated as the number of days between each customer's most recent purchase and a fixed reference date, set one day after the latest transaction recorded in the dataset. Frequency was calculated as the number of distinct invoices associated with each customer, thereby representing the number of separate purchasing occasions during the observation period. Monetary value was calculated as the sum of all line-level totals per customer, representing total spend over the observed period. This aggregation produced a dataset of 4,338 unique customers, each characterized by a Recency, Frequency, and Monetary value.

Descriptive statistics for the three RFM variables are presented in Table 2. Both Frequency and Monetary value show a mean that substantially exceeds the median, indicating right-skewed distributions in which a relatively small number of highly active, high-spending customers result in mean values that exceed those of the typical customer, as

represented by the median. This skew was confirmed visually through histograms of both variables, which showed the large majority of customers clustered at low values, with a long tail of high-frequency, high-spending customers extending to the right.

This distributional pattern had important methodological implications. The positive skew, combined with the presence of extreme values, motivated the use of Spearman's rank correlation in place of Pearson's correlation, and justified the subsequent log transformation of the Monetary variable prior to clustering. These methodological choices are described in Sections 3.4 and 3.5.

Table 2. Descriptive Statistics of RFM Variables

Variable	Minimum	Median	Mean	Maximum
Recency (days)	1	51	—	374
Frequency (orders)	1	2	4.27	209
Monetary (\$)	3.75	674	2,054	280,206

3.4 Correlation Analysis

A correlation analysis was conducted between the Frequency and Monetary variables to evaluate the strength and direction of their relationship. Given the right-skewed distributions of both variables and the presence of extreme values documented in Section 3.3, Spearman's rank correlation was selected over Pearson's product-moment correlation. Because Spearman's correlation is based on the rank order of observations rather than their raw values, it is less sensitive to skewed distributions and extreme observations than Pearson's product-moment correlation, which relies directly on the raw magnitude of each observation.

In addition to the Frequency-Monetary relationship, a full pairwise Spearman correlation matrix was computed across all three RFM variables — Recency, Frequency, and Monetary — to characterize the broader pattern of association among the dimensions of customer behavior. Correlation coefficients and associated significance tests were computed for each pair of variables.

3.5 Customer Segmentation

To address H2 — whether customers can be meaningfully segmented into behaviorally distinct groups based on the joint consideration of Frequency and Monetary value — k-means clustering was applied to the RFM dataset. Unlike correlation, which establishes whether a relationship exists between two variables, k-means clustering partitions observations into groups by minimizing the Euclidean distance between each observation and its assigned cluster centroid, allowing distinct behavioral segments to emerge from the joint structure of multiple variables simultaneously.

An initial clustering solution was explored using $k = 3$ as a preliminary specification. Because Monetary value showed substantial positive skew, as documented in Section 3.3, Monetary value was log-transformed prior to clustering, reducing the influence of extreme values and producing a distribution more suitable for Euclidean distance-based clustering. Both Frequency and log-transformed Monetary value were additionally scaled prior to clustering, ensuring that differences in the two variables' units and ranges did not disproportionately affect the resulting distance calculations.

The number of clusters, k , was selected using the elbow method, which examines the relationship between the number of clusters and within-cluster sum of squares to identify the point at which additional clusters yield diminishing reductions in within-cluster variance. This method was used to guide the selection of the final number of clusters. To ensure reproducibility and reduce sensitivity to random initialization, the final k-means model was estimated using a fixed random seed (123) and 25 random starting configurations ($nstart = 25$).

3.6 Validation of the Clustering Solution

Because the elbow method involves a degree of visual judgment rather than a definitive statistical criterion, silhouette analysis was conducted as a complementary validation technique to assess the appropriateness of the selected number of clusters. The silhouette coefficient measures how similar each observation is to its own cluster compared to the next-nearest cluster, producing an average silhouette width for each candidate value of k ; larger average silhouette widths indicate more clearly separated, internally cohesive clusters. Average silhouette width was computed across a range of candidate values ($k = 2$ through $k = 10$) to evaluate whether the previously selected $k = 4$ solution was well supported relative to alternative cluster counts.

3.7 Principal Component Analysis

To visualize the structure of the resulting customer segments in a reduced number of dimensions, principal component analysis (PCA) was conducted across all three RFM variables, using the same standardized variables employed in the k-means clustering (Section 3.5), ensuring consistency between the clustering and visualization procedures. PCA transforms a set of correlated variables into a smaller number of orthogonal principal components, ordered according to the proportion of total variance each component explains. The first two principal components were used to visualize the four customer segments identified through k-means clustering in two dimensions, providing a complementary, variance-based representation of the data that facilitated interpretation of the clustering solution. Because the visualization is derived from the underlying variance structure rather than the k-means optimization criterion itself, it helps assess whether the identified clusters correspond to genuine patterns in the data rather than artifacts of the clustering procedure.

4. Results

4.1 Correlation Results

The Spearman correlation between Frequency and Monetary value was strong and positive ($\rho = 0.807$, $p < .001$), indicating that customers who purchase more frequently also tend to spend substantially more overall. This result is consistent with H0's characterization of Frequency and Monetary value as strongly related, rather than H1's prediction that the two variables would be largely independent. It is worth noting, however, that a strong correlation demonstrates association between the two variables, not that they measure an identical underlying construct; the cluster analysis represented in Section 4.2 provides a complementary evaluation of whether Frequency and Monetary value nonetheless behave as distinct dimensions when customers are grouped jointly on both.

The full pairwise correlation matrix across all three RFM variables is presented in Table 3. In addition to the strong positive Frequency-Monetary relationship, Recency was negatively correlated with both Frequency ($\rho = -0.563$) and Monetary value ($\rho = -0.483$), indicating that customers with lower Recency values — that is, more recent purchases — also tend to purchase more frequently and spend more overall. All reported correlations were statistically significant.

Table 3. Spearman Correlation Matrix of RFM Variables

	Recency	Frequency	Monetary
Recency	1.000	-0.563	-0.483
Frequency	-0.563	1.000	0.807
Monetary	-0.483	0.807	1.000

Note. Values are Spearman rank correlation coefficients. All correlations are statistically significant, $p < .001$.

4.2 Clustering Results

The initial $k = 3$ clustering solution, conducted on the untransformed Frequency and Monetary variables, produced three clusters of highly unequal size: 4,095 customers in one cluster (mean Frequency = 3.07 orders, mean Monetary = \$1,105), 230 customers in a second (mean Frequency = 21.3 orders, mean Monetary = \$11,882), and only 13 customers in a third (mean Frequency = 82.5 orders, mean Monetary = \$127,338). This imbalance is consistent with the extreme right-skew in Monetary value documented in Section 3.3, in which a small number of high-spending customers disproportionately influenced cluster assignment on the untransformed scale.

Following the log transformation and scaling described in Section 3.5, the elbow analysis supported the selection of $k = 4$ as an appropriate balance between reducing within-cluster variance and maintaining an interpretable segmentation. The final $k = 4$ solution produced four behaviorally distinct customer segments, summarized in Table 4. The largest segment, comprising 2,075 customers, consisted of rare or one-time buyers, averaging approximately 1.49 orders and \$307 in total spend. A second segment of 1,939 customers represented regular repeat customers, averaging

approximately 4.42 orders and \$1,768 in total spend. A smaller segment of 316 customers represented loyal, high-value customers, averaging approximately 18.6 orders and \$13,301 in total spend. Finally, a very small segment of 8 customers represented extreme high-value outliers, averaging approximately 122 orders and \$80,210 in total spend.

The gap in Monetary value between the regular repeat segment and the loyal high-value segment was proportionally larger than the corresponding gap in Frequency, indicating that Frequency alone would understate the difference in customer value between these two groups. This finding underscores the value of segmenting on Frequency and Monetary value jointly, rather than relying on either dimension in isolation, even in a context where the two variables are strongly correlated overall.

Because the final clustering analysis was performed using the standardized Frequency and log-transformed Monetary variables, Table 4 reports the corresponding cluster means for those two variables. Recency was retained as a descriptive RFM measure and included in the correlation analysis but was not used in the clustering procedure.

Table 4. Final K-Means Cluster Solution (k = 4)

Segment	n	Mean Frequency (orders)	Mean Monetary (\$)
Rare / one-time buyers	2,075	1.49	307
Regular repeat customers	1,939	4.42	1,768
Loyal, high-value customers	316	18.6	13,301
Extreme high-value outliers	8	122	80,210

4.3 Validation of the Clustering Solution

Average silhouette widths were computed for candidate clustering solutions ranging from k = 2 to k = 10 (Table 5). The highest average silhouette width was observed for k = 2 (0.522), with average silhouette width declining gradually as the number of clusters increased, reaching 0.410 at k = 10. The four-cluster solution produced an average silhouette width of 0.504, only modestly below the maximum value observed for k = 2.

Although the highest average silhouette width was observed for k = 2, k = 4 was retained for the final analysis. A two-cluster solution, while exhibiting slightly greater internal cohesion according to the silhouette criterion, would merge customer groups that differ substantially in purchasing behavior. In particular, it would eliminate the distinction between regular repeat customers and loyal, high-value customers established in Section 4.2, thereby reducing the behavioral resolution of the segmentation. The relatively small gap between the silhouette widths of k = 2 and k = 4 (0.522 vs. 0.504) suggests that k = 4 remains a well-supported and defensible choice, offering substantially greater behavioral interpretability while incurring only a modest reduction in average cluster cohesion.

Table 5. Average Silhouette Width by Number of Clusters (k)

k	Average Silhouette Width
2	0.522
3	0.520
4	0.504
5	0.470
6	0.458
7	0.441
8	0.435
9	0.412
10	0.410

4.4 Principal Component Analysis

Principal component analysis was performed using the same standardized variables employed in the final k-means clustering solution. The first principal component (PC1) accounted for 61.85% of total variance, and the second principal component (PC2) accounted for an additional 25.04%, together explaining 86.89% of the total variance in the standardized data.

When the four customer segments identified in Section 4.2 were plotted against these two components, the four customer segments exhibited their greatest separation along the first principal component (PC1), with comparatively less separation along PC2. The rare/one-time buyer and regular repeat customer segments occupied the lower range of PC1, with substantial overlap between them, while the loyal, high-value and extreme high-value outlier segments occupied progressively higher positions along PC1, with the extreme outlier segment positioned furthest from the rest of the customer base on both components. This gradual progression, rather than the presence of four sharply separated groups, is consistent with customer purchasing behavior varying along a continuous spectrum rather than forming naturally discrete categories. The k-means algorithm therefore partitions this continuous behavioral space into interpretable customer segments that are useful for analysis and managerial decision-making.

5. Discussion

The findings of this study provide insight into the research question motivating the analysis: whether purchase frequency and monetary value represent a single underlying dimension of consumer behavior or two behavioral dimensions that together reveal meaningful customer segments. The strong positive correlation between Frequency and Monetary value ($\rho = 0.807$) indicates that, on average, these two variables move together closely, supporting H0's characterization of a strong overall relationship. However, the clustering results demonstrate that this strong average

relationship does not preclude meaningful behavioral heterogeneity. The four-segment solution — rare/one-time buyers, regular repeat customers, loyal high-value customers, and extreme outliers — showed that the gap in Monetary value between the regular repeat and loyal high-value segments was proportionally larger than the corresponding gap in Frequency, meaning a segmentation based on Frequency alone would understate genuine differences in customer value. Taken together, these findings suggest that Frequency and Monetary value, while strongly correlated in the aggregate, retain sufficient unique information to justify treating them jointly rather than substituting one for the other — a conclusion directly relevant to the tension identified in the Introduction between practitioner convenience and behavioral precision.

The principal component analysis reinforces this interpretation. Rather than revealing four sharply bounded groups, the PCA visualization showed customer segments arranged along a continuous gradient, primarily along the first principal component. This is consistent with purchasing behavior varying continuously across the customer base, and it suggests that the value of the k-means solution lies not in discovering naturally occurring, discrete customer types, but in partitioning a continuous spectrum of customer purchasing behavior into segments that are interpretable and actionable for marketing decision-making.

From a practical standpoint, these findings suggest that treating high-frequency and high-monetary-value customers as interchangeable would risk overlooking a meaningful minority of customers whose value is disproportionate to how often they purchase. Marketing resource allocation — for example, retention efforts or loyalty incentives — may benefit from accounting for both dimensions jointly rather than relying on frequency as a proxy for overall value. This distinction may be particularly important for customer lifetime value strategies, where purchase frequency alone may fail to identify customers whose infrequent purchases nevertheless generate substantial revenue.

Several limitations should be noted when interpreting these results. Because customer-level segmentation requires transactions to be linked to an individual customer, approximately 25% of transactions — those without a recorded Customer ID, likely representing guest checkouts or otherwise unidentified customers — could not be included in the analysis. This means the segmentation reflects only identified, repeat-eligible customers, and the observed segment structure may not generalize to guest-checkout behavior, which could plausibly differ in composition and value. Retailers seeking to apply this type of segmentation would benefit from capturing a persistent identifier for guest transactions (for example, linked by email) to allow this population to be studied as well. Additionally, this analysis draws on a single UK-based online retailer over a two-year period; the specific segment boundaries and cluster sizes observed here may not generalize to other retail contexts, product categories, or time periods. Finally, k-means clustering assumes roughly spherical, similarly sized clusters in the underlying variable space, an assumption that may not perfectly hold for the behavioral data examined here, even after log-transformation and scaling.

Despite these limitations, the consistency of the correlation, clustering, silhouette, and PCA analyses suggests that the identified customer segments capture meaningful behavioral differences within this retail dataset, providing a useful foundation for future research on customer segmentation using transactional data.

6. Conclusion

This study investigated whether purchase frequency and monetary value represent a single underlying dimension of customer purchasing behavior or two distinct behavioral dimensions that together reveal meaningful customer segments. Using RFM analysis and k-means clustering on transaction data from an online UK-based retailer, the analysis found that while Frequency and Monetary value are strongly correlated overall, segmenting customers on both dimensions jointly reveals behavioral distinctions — most notably between regular repeat customers and loyal, high-value customers — that a single-dimension view would obscure. Validation through silhouette analysis and principal component analysis supported the retained four-segment solution as a defensible and interpretable representation of the underlying customer base, while also demonstrating that customer purchasing behavior is better represented as a continuous spectrum than as a set of naturally discrete groups.

Beyond its specific findings, this study illustrates a broader methodological point relevant to consumer behavior research: a strong correlation between two behavioral measures does not, by itself, justify collapsing them into a single construct. Complementary analytical techniques — here, clustering and dimensionality reduction alongside correlation — can reveal structure that a single method would miss, even when that structure exists within variables that are themselves highly correlated. Future research could extend this analytical framework to additional retail contexts, incorporate temporal dynamics in purchasing behavior, and examine whether the customer segments identified here differ in their responses to targeted marketing interventions.

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Appendix

A. Reproducibility

The R session information below records the exact R version, operating system, and package versions used for this analysis, allowing the full workflow to be reproduced.

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.2
```

```
attached base packages:
stats, graphics, grDevices, utils, datasets, methods, base
```

```
other attached packages:
cluster_2.1.8.1, purrr_1.1.0, ggplot2_4.0.0, lubridate_1.9.4, dplyr_1.1.4
```

B. Data Cleaning Code

```
library(readr)
library(dplyr)
library(lubridate)

data.df <- read_csv("data/online_retail_II.csv")

clean.df <- data.df %>%
  filter(!grepl("^C", Invoice)) %>%      # remove cancelled orders
  filter(Price > 0) %>%                 # remove non-positive prices
  filter(!is.na(Customer.ID)) %>%      # remove missing customer IDs
  mutate(InvoiceDate = mdy_hm(InvoiceDate)) # convert text to real date-time
```

C. RFM Construction & Correlation Code

```
# Build RFM table
reference_date <- max(clean.df$InvoiceDate) + days(1)

rfm.df <- clean.df %>%
  mutate(LineTotal = Quantity * Price) %>%
  group_by(Customer.ID) %>%
  summarise(
    Recency    = as.numeric(difftime(reference_date, max(InvoiceDate), units =
"days")),
    Frequency  = n_distinct(Invoice),
    Monetary   = sum(LineTotal)
  )

# Spearman correlation: Frequency vs Monetary
cor_result <- cor.test(rfm.df$Frequency, rfm.df$Monetary, method = "spearman")
cor_result
```



```
# Full RFM correlation matrix
rfm_matrix <- rfm.df %>%
  select(Recency, Frequency, Monetary) %>%
  cor(method = "spearman")
print(rfm_matrix)
```

D. Clustering, Silhouette & PCA Code

```
# Log-transform Monetary to address skew
rfm.df <- rfm.df %>%
  mutate(Log_Monetary = log(Monetary + 1))

# Scale variables for clustering
scaled.df <- rfm.df %>%
  select(Frequency, Log_Monetary) %>%
  scale()

# Elbow method
library(purrr)
set.seed(123)
wss <- map_dbl(1:10, function(k) {
  kmeans(scaled.df, centers = k, nstart = 25)$tot.withinss
})

# Final k-means clustering (k = 4)
set.seed(123)
kmeans_result_k4 <- kmeans(scaled.df, centers = 4, nstart = 25)
rfm.df$Cluster_k4 <- as.factor(kmeans_result_k4$cluster)

# Silhouette analysis (k = 2 through 10)
library(cluster)
sil_widths <- map_dbl(2:10, function(k) {
  km <- kmeans(scaled.df, centers = k, nstart = 25)
  ss <- silhouette(km$cluster, dist(scaled.df))
  mean(ss[, 3])
})

# Principal component analysis (Recency, Frequency, Log_Monetary)
pca_input <- rfm.df %>%
  select(Recency, Frequency, Log_Monetary) %>%
  scale()
pca_result <- prcomp(pca_input, center = TRUE, scale. = TRUE)
summary(pca_result)
```

E. Figures

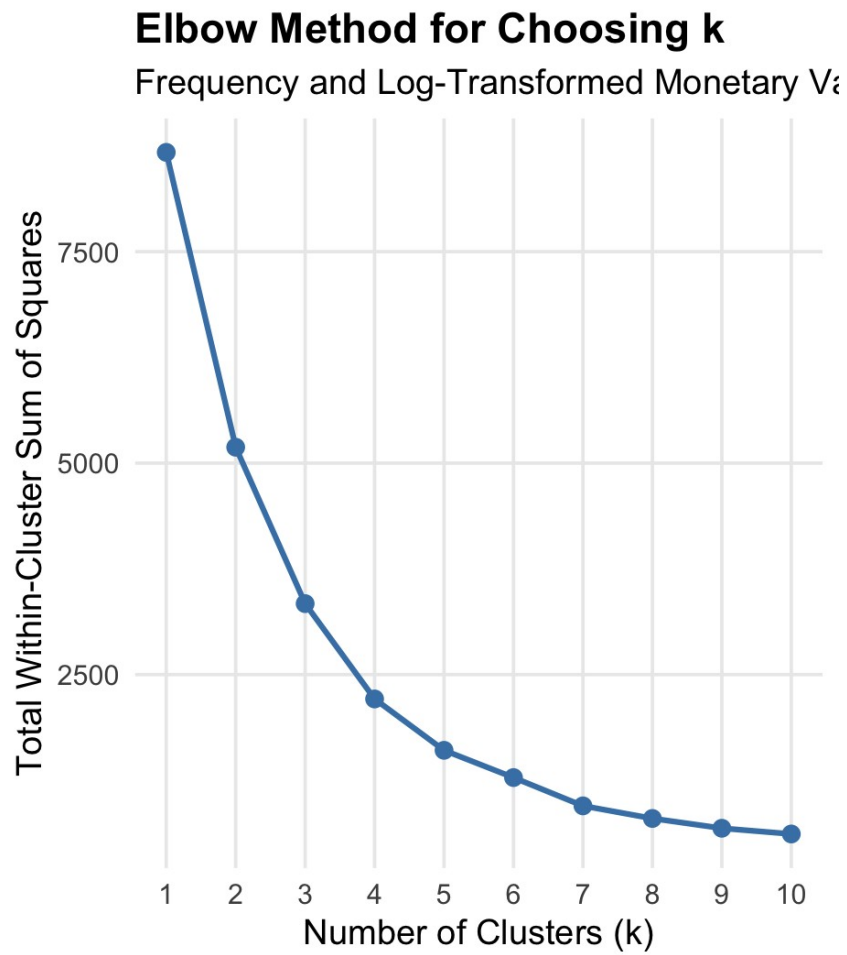


Figure 1. Elbow method for choosing k (total within-cluster sum of squares by number of clusters, $k = 1-10$).

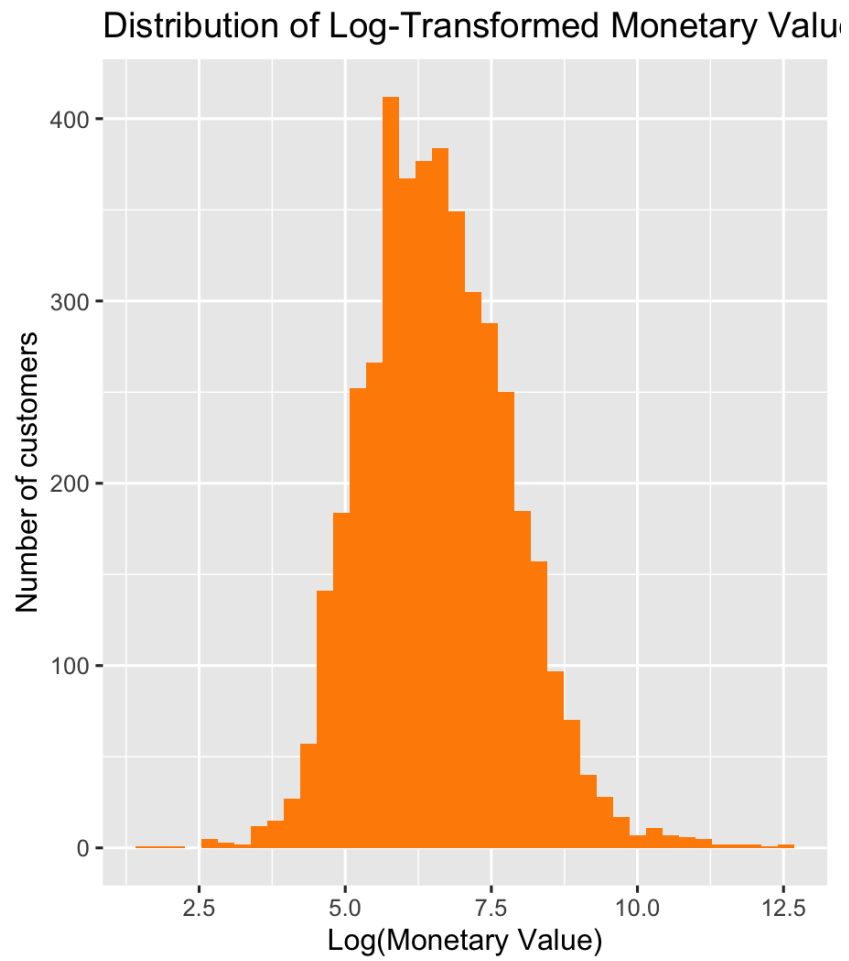


Figure 2. Distribution of log-transformed Monetary value, following the transformation described in Section 3.5.

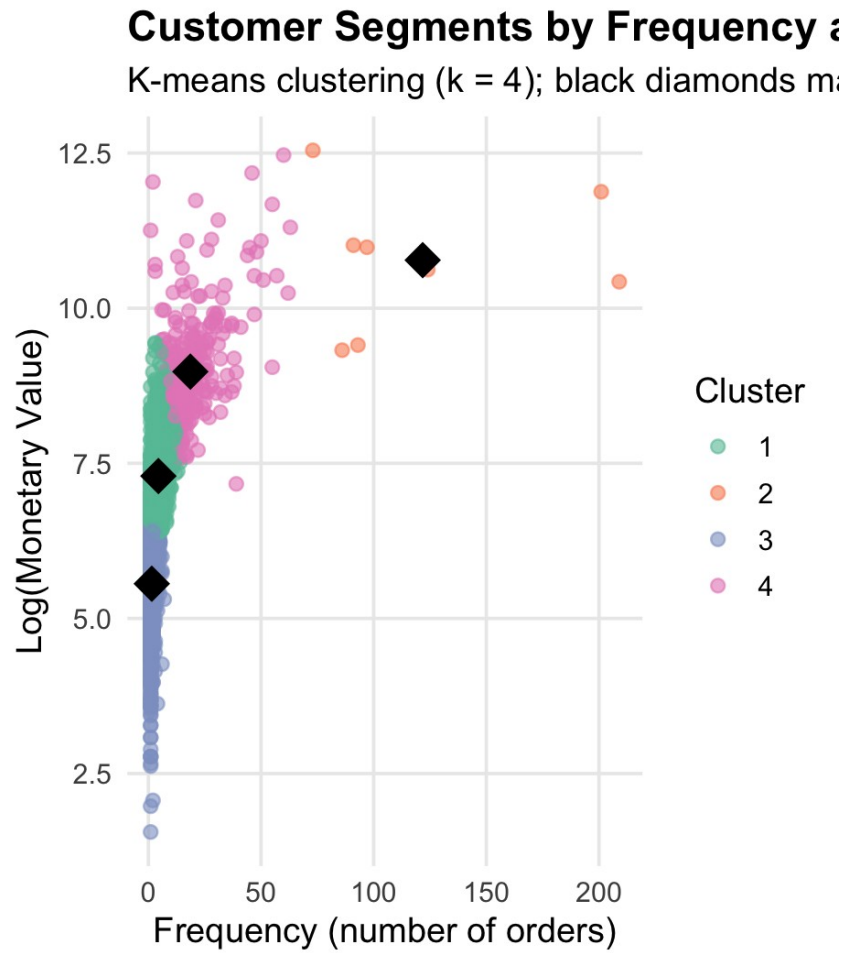


Figure 3. Final k -means cluster solution ($k = 4$): customers plotted by Frequency and log-transformed Monetary value, with cluster centroids marked by black diamonds.

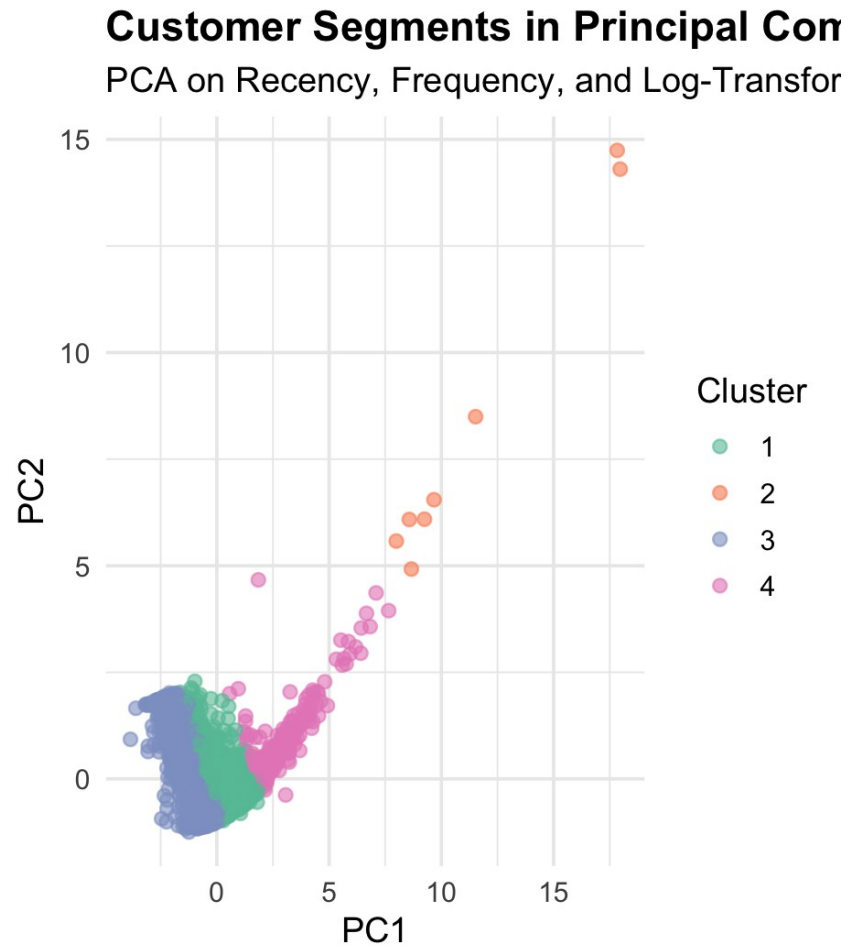


Figure 4. Customer segments projected into principal component space (PC1 vs. PC2), colored by the $k = 4$ cluster solution.